WORKED SOLUTION

STEP I 1995, Question 3

Pure • Telescoping Series

> **STEP I 1995, Question 3**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/95-s1-q3

The key idea

When consecutive terms of a sum share most of their parts, writing each term as a difference $\mathrm{f}(r) - \mathrm{f}(r-1)$ makes almost everything cancel: the sum *telescopes*. The work is in choosing f so that the difference is the term you want; here the question chooses it for us. For the alternating series, the trick is to write it as *all* the cubes minus twice the even ones, so that part (ii) does the work twice.

Solution

Part (i)

Write the sum out in full:

$$\sum_{r=1}^{n} \{\mathrm{f}(r) - \mathrm{f}(r-1)\} = \{\mathrm{f}(1) - \mathrm{f}(0)\} + \{\mathrm{f}(2) - \mathrm{f}(1)\} + \cdots + \{\mathrm{f}(n) - \mathrm{f}(n-1)\}.$$

Each of $\mathrm{f}(1), \mathrm{f}(2), \ldots, \mathrm{f}(n-1)$ appears once with a plus sign and once with a minus sign, so they all cancel. Only $-\mathrm{f}(0)$ from the first bracket and $\mathrm{f}(n)$ from the last survive, giving $\mathrm{f}(n) - \mathrm{f}(0)$, as required.

If you'd like something more formal than "everything cancels", use induction on n. For $n = 1$ the sum is $\mathrm{f}(1) - \mathrm{f}(0)$. If the result holds for $n = k$, then adding the next term gives

$$\mathrm{f}(k) - \mathrm{f}(0) + \mathrm{f}(k+1) - \mathrm{f}(k) = \mathrm{f}(k+1) - \mathrm{f}(0),$$

which is the result for $n = k+1$. So it holds for all $n \geq 1$.

Part (ii)

With $f(r) = r^2(r+1)^2$ we have $f(r-1) = (r-1)^2r^2$. Both contain the factor r^2, so take it out and use a difference of two squares:

$$\begin{aligned} f(r) - f(r-1) &= r^2[(r+1)^2 - (r-1)^2] \\ &= r^2[(r+1) + (r-1)][(r+1) - (r-1)] \\ &= r^2 \cdot 2r \cdot 2 = 4r^3. \end{aligned}$$

Now sum from $r = 1$ to n and apply part (i). Since $f(0) = 0$,

$$4\sum_{r=1}^{n} r^3 = f(n) - f(0) = n^2(n+1)^2, \qquad \text{so} \qquad \boxed{\sum_{r=1}^{n} r^3 = \tfrac{1}{4}n^2(n+1)^2.}$$

Part (iii)

The series has $2n+1$ terms: odd cubes with a plus sign and even cubes with a minus sign, ending on the odd cube $(2n+1)^3$.

Ask yourself How can we deal with the alternating signs using only the formula for $\sum r^3$?

Start from the sum of *all* the cubes up to $(2n+1)^3$. That has every even cube with a plus sign, where we want a minus sign, so subtract each even cube twice:

$$1^3 - 2^3 + 3^3 - \cdots + (2n+1)^3 = \sum_{k=1}^{2n+1} k^3 - 2\sum_{m=1}^{n} (2m)^3.$$

The even cubes are $(2m)^3 = 8m^3$ for $m = 1, \ldots, n$, the largest being $(2n)^3$. Using part (ii) twice, once with $2n+1$ in place of n,

$$\begin{aligned} \sum_{k=1}^{2n+1} k^3 - 16\sum_{m=1}^{n} m^3 &= \tfrac{1}{4}(2n+1)^2(2n+2)^2 - 16 \cdot \tfrac{1}{4}n^2(n+1)^2 \\ &= (2n+1)^2(n+1)^2 - 4n^2(n+1)^2 \\ &= (n+1)^2[(2n+1)^2 - (2n)^2] \\ &= (n+1)^2(4n+1), \end{aligned}$$

where in the second line we used $(2n+2)^2 = 4(n+1)^2$, and in the last the difference of two squares $(2n+1)^2 - (2n)^2 = (2n+1+2n) \cdot 1$. So

$$\boxed{1^3 - 2^3 + 3^3 - 4^3 + \cdots + (2n+1)^3 = (n+1)^2(4n+1).}$$

Checks

- For $n = 0$ the series is just $1^3 = 1$, and $(0+1)^2(0+1) = 1$.

- For $n = 1$: $1 - 8 + 27 = 20$, and $2^2 \times 5 = 20$.
- For $n = 2$: $1 - 8 + 27 - 64 + 125 = 81$, and $3^2 \times 9 = 81$.
- The answer should grow like $(2n+1)^3/2$, roughly half the last term, since the terms pair up as $(2m+1)^3 - (2m)^3 \approx 12m^2$. Indeed $(n+1)^2(4n+1) \approx 4n^3$, matching $\sum 12m^2 \approx 4n^3$.

We also checked parts (ii) and (iii) by direct summation for every n from 0 to 199.

Common slips

- Subtracting the even cubes only once, which gives $\sum k^3 - \sum (2m)^3$, the sum of the *odd* cubes, not the alternating sum.
- Taking the even cubes up to $(2n+2)^3$ or using n in place of $2n+1$ in the formula for all the cubes: count the terms carefully.

Beyond STEP

Part (i) is the discrete analogue of the fundamental theorem of calculus: the difference $\mathrm{f}(r) - \mathrm{f}(r-1)$ plays the role of the derivative, and summing it recovers f, just as integrating F' gives $F(b) - F(a)$. This "calculus of finite differences" lets you sum any polynomial by finding a suitable f, and the same telescoping idea underlies many convergence proofs for series. Siklos, *Advanced Problems in Mathematics*, has more practice with telescoping and series of this kind.

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